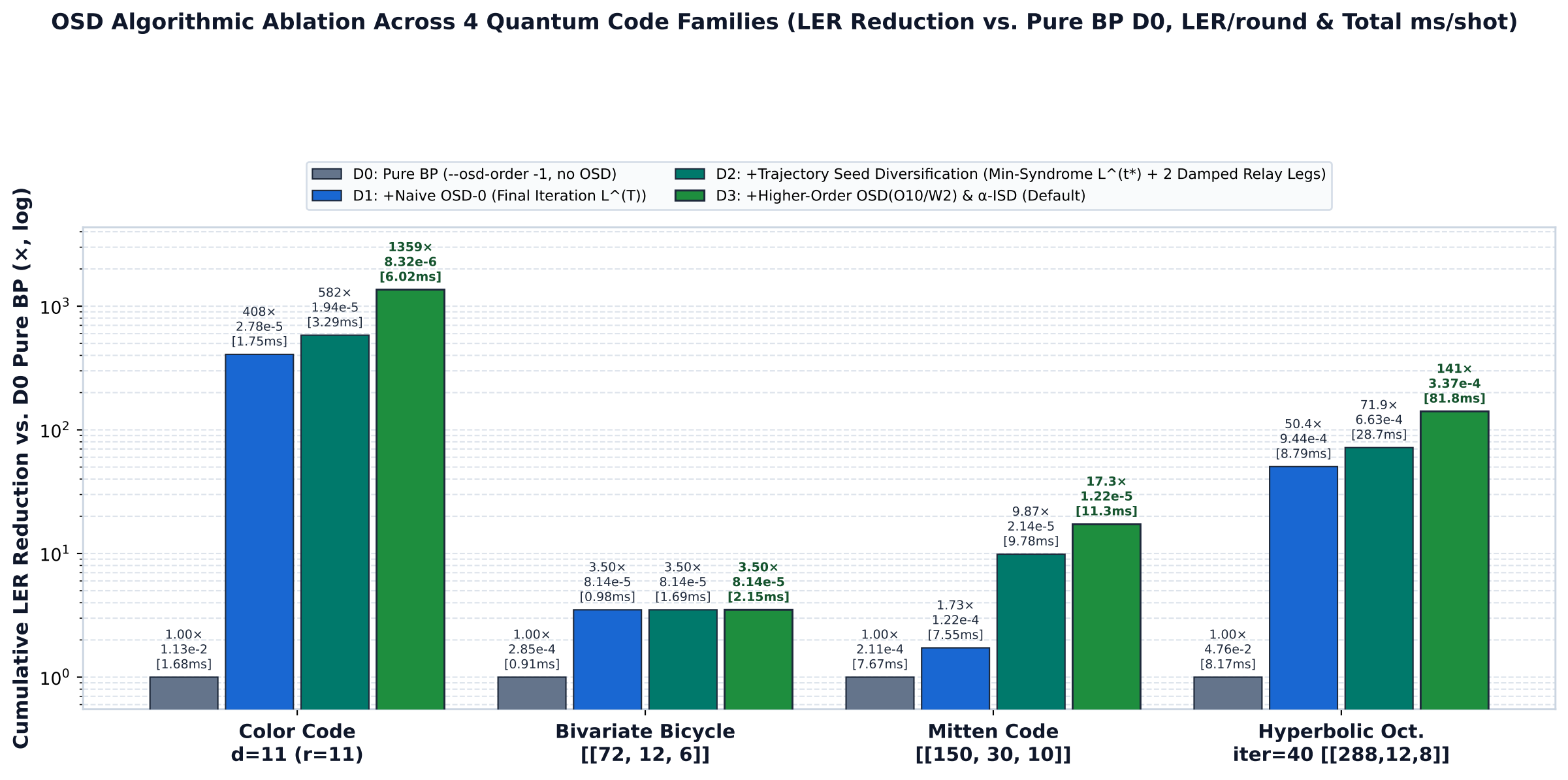
OSD Algorithmic Ablation Across 4 Quantum Code Families (LER Reduction vs. Pure BP D0, LER/round & Total ms/shot)
D0: Pure BP (--osd-order -1, no OSD)
D1: +Naive OSD-0 (Final Iteration L^(T))
D2: +Trajectory Seed Diversification (Min-Syndrome L^(t*) + 2 Damped Relay Legs)
D3: +Higher-Order OSD(O10/W2) & α-ISD (Default)
Cumulative LER Reduction vs. D0 Pure BP (×, log)
10^0
10^1
10^2
10^3
1.00× 1.13e-2 [1.68ms]
408× 2.78e-5 [1.75ms]
582× 1.94e-5 [3.29ms]
1359× 8.32e-6 [6.02ms]
1.00× 2.85e-4 [0.91ms]
3.50× 8.14e-5 [0.98ms]
3.50× 8.14e-5 [1.69ms]
3.50× 8.14e-5 [2.15ms]
1.00× 2.11e-4 [7.67ms]
1.73× 1.22e-4 [7.55ms]
9.87× 2.14e-5 [9.78ms]
17.3× 1.22e-5 [11.3ms]
1.00× 4.76e-2 [8.17ms]
50.4× 9.44e-4 [8.79ms]
71.9× 6.63e-4 [28.7ms]
141× 3.37e-4 [81.8ms]
Color Code d=11 (r=11)
Bivariate Bicycle [[72, 12, 6]]
Mitten Code [[150, 30, 10]]
Hyperbolic Oct. iter=40 [[288,12,8]]